

Ortogonalización

- C6S
- M6S

Householder (Reflexion) Rápido

Givens (Rotacion) Rápido

$$A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{bmatrix}$$

$a_i \in \mathbb{R}^m$

↓
L.I.

$$A \stackrel{?}{=} \begin{matrix} Q & R \\ m \times n & m \times m \end{matrix}$$

$$G^T(i, k, \theta) = \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & \ddots & & & \\ & & & c & & -s \\ & \ominus & & s & & \\ & & & & \ddots & \\ & & & & & c \\ & & \ominus & & & s \\ & & & & & & 1 \end{pmatrix}$$

Rotacion de θ en plano i, k .

$$G^T \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_k \end{pmatrix} = \begin{matrix} i \\ k \end{matrix} \begin{pmatrix} r_q \\ c r_i - s r_k \\ s r_i + c r_k \\ r_e \end{pmatrix}$$

$$A = \begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

$$i=1 \\ k=2$$

$$\begin{pmatrix} d_1^2 & \\ & d_2^2 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

$$= G_3 G_2 \begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

$$\begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

$$i=1 \\ k=4$$

$$= G_3 \begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

$$= \begin{pmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{pmatrix}$$

$$G_N \dots G_2 G_1 A = R$$

$$A = QR$$

$$Q^T = \dots G_2 G_1$$

$$Q = \begin{bmatrix} G_1^T & G_2^T & \dots & G_N^T \end{bmatrix}$$

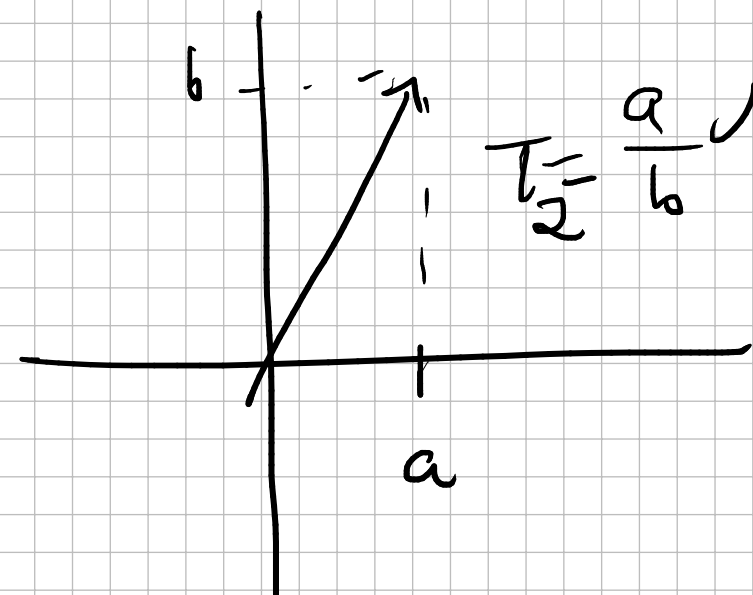
$$G_2 G_1 = \begin{pmatrix} c_1 & -s_1 \\ s_1 & c_1 \end{pmatrix} = \begin{pmatrix} -s_1 & 0 \\ 0 & s_1 \end{pmatrix} \begin{pmatrix} -t_1 & 1 \\ 1 & t_1 \end{pmatrix}$$

$$t = -\frac{c}{s}$$

$$G_2 G_1 = \begin{pmatrix} -s_2 & 0 \\ 0 & +s_2 \end{pmatrix} \begin{pmatrix} -t_2 & 1 \\ 1 & t_2 \end{pmatrix} \begin{pmatrix} -s_1 & 0 \\ 0 & s_1 \end{pmatrix} \begin{pmatrix} -t_1 & 1 \\ 1 & t_1 \end{pmatrix}$$

$$= \begin{pmatrix} -s_2 & 0 \\ 0 & s_2 \end{pmatrix} \begin{pmatrix} d_2 & 0 \\ 0 & d_1 \end{pmatrix} \begin{pmatrix} -\left(\frac{d_1}{d_2}\right)^2 t_2 & 1 \\ 1 & t_2 \end{pmatrix} \begin{pmatrix} -t_1 & 1 \\ 1 & t_1 \end{pmatrix}$$

Diagonal $\underbrace{S_N \dots S_2 S_1}_M$ $\underbrace{T_N \dots T_2 T_1}_R$ $A =$ triangular sum



$$t_2 = \frac{d_1 a}{d_2 b}$$

$$\begin{pmatrix} * \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Dependencia Lineal. HH, 6.

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix}$$

Aplicans H.H.

matriz de reemplazo

$$\begin{pmatrix} * & * \\ 0 & 0 \\ 0 & \vdots \\ 0 & 0 \end{pmatrix}$$

$$H_r \dots H_2 H_1 A \Pi_1 \Pi_2 \dots \Pi_r = \begin{pmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{pmatrix}$$

$$\begin{matrix} n-r \\ \hline r \end{matrix}$$

si rango(A) = r, $R_{22} = 0$

$$R_{1L} \rightsquigarrow 0$$

$$\begin{pmatrix} R_{11} & R_{12} \end{pmatrix}^T = \begin{pmatrix} R_{11}^T \\ R_{12}^T \end{pmatrix} = \begin{pmatrix} \text{triangular superior} \\ \text{triangular inferior} \end{pmatrix}$$

$$\underbrace{Z_r \dots Z_1}_{Q^T} \begin{pmatrix} R_{11}^T \\ R_{12}^T \end{pmatrix} = \begin{pmatrix} \text{triangular superior} \\ 0 \end{pmatrix} \xrightarrow{T=T^T} \begin{pmatrix} R_{11} & R_{12} \end{pmatrix} Q^T = \begin{pmatrix} \text{triangular inferior} & 0 \end{pmatrix}$$

$\Pi_2 =$ Resulta de Insn permut column

$$\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} a_1 & c_1 & b_1 \\ a_2 & c_2 & b_2 \\ a_3 & c_3 & b_3 \end{pmatrix}$$

$$\underbrace{H_r \dots H_2 H_1}_{Q^T} A \underbrace{\pi_1 \pi_2 \dots \pi_r}_{\Pi} = \begin{pmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{pmatrix}$$

$\begin{matrix} \text{---} r \text{---} & \text{---} r \text{---} & \text{---} n-r \text{---} \\ \text{---} r \text{---} & \text{---} r \text{---} & \text{---} n-r \text{---} \end{matrix}$

$$\begin{pmatrix} R_{11} & R_{12} \end{pmatrix} \tilde{Q} = \begin{pmatrix} T_{11} & 0 \end{pmatrix}$$

$\begin{matrix} \text{---} r \text{---} \\ \text{---} n \text{---} \end{matrix}$

$$\overbrace{Q^T A \Pi \tilde{Q}}^{m \times m \quad m \times n \quad n \times n \quad n \times n} = \begin{pmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{pmatrix} \tilde{Q} = \begin{pmatrix} T_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

$\begin{matrix} \text{---} T.S \text{---} \\ \text{---} T.S \text{---} \end{matrix}$

$$\tilde{\pi}_1 = \begin{pmatrix} 0 & \vdots & 0 \\ \vdots & 0 & \\ 0 & & I \end{pmatrix}$$

$$\tilde{\pi}_2 = \begin{pmatrix} \vdots & 0 \\ 0 & I \end{pmatrix}$$

Q is complete

$$\underbrace{\tilde{\pi}_2 \tilde{Q}^T A \Pi \tilde{Q} \pi_1}_{\tilde{Q}^T} = \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

$$\leftarrow \tilde{Q}^T A \tilde{Z} = \begin{pmatrix} \tilde{A} & 0 \\ 0 & 0 \end{pmatrix}$$

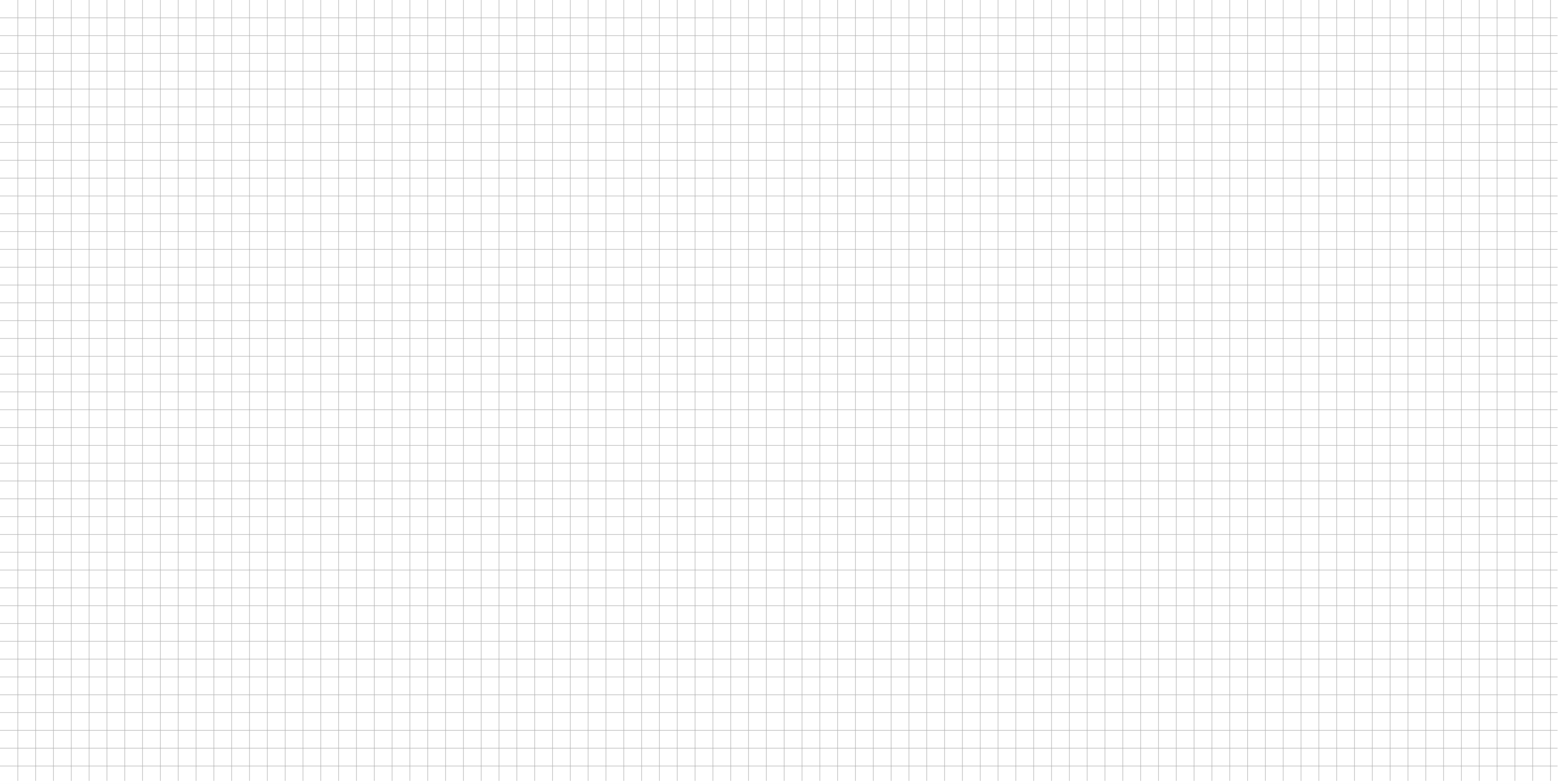
$$\underbrace{Q^T}_{m \times m} \underbrace{A}_{m \times n} \underbrace{Z}_{n \times n} = \begin{pmatrix} \text{diag} & 0 \\ 0 & 0 \end{pmatrix}$$

Note:

$$\underbrace{Q^T A Q}_{m \times m} = \begin{pmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{pmatrix}$$

$$\begin{bmatrix} Q_1 & Q_2 \end{bmatrix}_{r \times r}^T A \begin{bmatrix} z_1 & z_2 \end{bmatrix}_{r \times 1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

R-diagonalization



Bidiagonalization

$$\begin{array}{c}
 H_1 B \\
 \downarrow \\
 H_2 \underbrace{H_1}_{\text{blue circle}} B \overset{22}{H_2} \overset{22}{H_3} \\
 \downarrow \\
 \underbrace{H_n \dots H_1}_{Q^T} B \underbrace{\overset{22}{H_2} \dots H_n}_{\tilde{Q}} = \begin{pmatrix} \diagup & & 0 \\ & \diagdown & \\ 0 & & \diagup \end{pmatrix}
 \end{array}$$

$$\begin{pmatrix} * & * & 0 & 0 \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

$$\begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{pmatrix}$$

$\boxed{* \ 0 \ 0}$
 H_2
 H_1
 $H_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & & \\ 0 & & 2 & \\ 0 & & & H_1 \end{pmatrix}$

Minimums cuadrados.

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix} \quad a_i \in \mathbb{R}^m$$

$$A = Q \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix} Z^T =$$

$$\begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R_{11} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 & z_2 \end{bmatrix}^T$$

$$\begin{matrix} m \times m & m \times n & n \times n \end{matrix} \quad Q^T A Z = \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix}$$

no singular.

$$Q_1^T R_{11} z_1$$

$$\text{Imagen}(A) = \underline{\text{span } Q_1}$$

$$\text{Imagen}(A)^\perp = \text{span } Q_2$$

$$\boxed{\text{Ker } A = \text{span } z_2}$$

$$\underline{\text{Ker } A^\perp = \text{span } z_1}$$

$$A = Q_1 R_{11} z_1^T$$

$$Q_1^T A z_1^T = R_{11}$$

$$A x = b$$

$m \times n$

$$b \in \mathbb{R}^m$$

$$x = ? \quad \underline{x \in \mathbb{R}^n}$$

$$A = Q_1^T R_{11} Z_1$$

$$A^{-1} \stackrel{?}{=} Z_1 R_{11}^{-1} Q_1^T$$

$b \notin \text{image}(A) \rightarrow$ project b on $\text{Image } A$.

$$\underline{P_{\text{Image}(A)}} b = \underline{P_{\text{span } Q_1}} b = \underline{Q_1 Q_1^T b} \in \text{Image } A$$

$$A x = Q_1 Q_1^T b$$

$$x = \text{span } Z_1 \rightarrow x = Z_1 \alpha, \alpha \in \mathbb{R}^r$$

$$\underline{Q_1^T A Z_1} \alpha = \underline{Q_1^T Q_1 Q_1^T b}$$

$$R_{11} \alpha = Q_1^T b$$

$$\alpha = R_{11}^{-1} Q_1^T b$$

$$x = \underline{Z_1 R_{11}^{-1} Q_1^T b}$$

$$Pb = \underbrace{(b^T q_1)}_{q_1^T b} q_1 + \underbrace{(b^T q_2)}_{q_2^T b} q_2 = \underline{Q Q^T b}$$

$$Q = \begin{bmatrix} q_1 & q_2 \\ 1 & 1 \end{bmatrix}$$

$$Q^T b = \begin{bmatrix} q_1^T b \\ q_2^T b \end{bmatrix}$$

$$\|Ax^* - b\|_2 = \min_{x \in \mathbb{R}^n} \|Ax - b\|_2$$

$$= \min_{x \in \mathbb{R}^n} \|\underline{Q}^T A z x - Q^T b\|_2$$

$$= \min_{x \in \mathbb{R}^n} \left\| \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}^T b \right\|_2$$

$$= \min_{\substack{x_1 \in \mathbb{R}^r \\ x_2 \in \mathbb{R}^{n-r}}} \left\| \begin{pmatrix} R_{11} & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{pmatrix} Q_1^T b \\ Q_2^T b \end{pmatrix} \right\|_2 = \min_{x_1 \in \mathbb{R}^r} \left\| \begin{pmatrix} R_{11} x_1 - Q_1^T b \\ -Q_2^T b \end{pmatrix} \right\|_2$$

$$= \min_{x_1 \in \mathbb{R}^r} \left(\|R_{11} x_1 - Q_1^T b\|^2 + \|Q_2^T b\|^2 \right)$$

$$\rightarrow x_1^* = R_{11}^{-1} Q_1^T b, \quad x^* = \begin{bmatrix} z_1 & z_2 \end{bmatrix} \begin{bmatrix} x_1^* \\ x_2 \end{bmatrix}$$

$$\|u\|_2 = \|Qu\|_2$$

$$(\underline{Qu})^T (Qu)$$

$$u^T \underline{Q^T Q} u$$

$$Z_{n \times n} x = x$$

$$x = \underline{Z^T} x$$

$$x \in \mathbb{R}^n$$

$$x = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \begin{matrix} 1 \times \\ n-r \end{matrix}$$

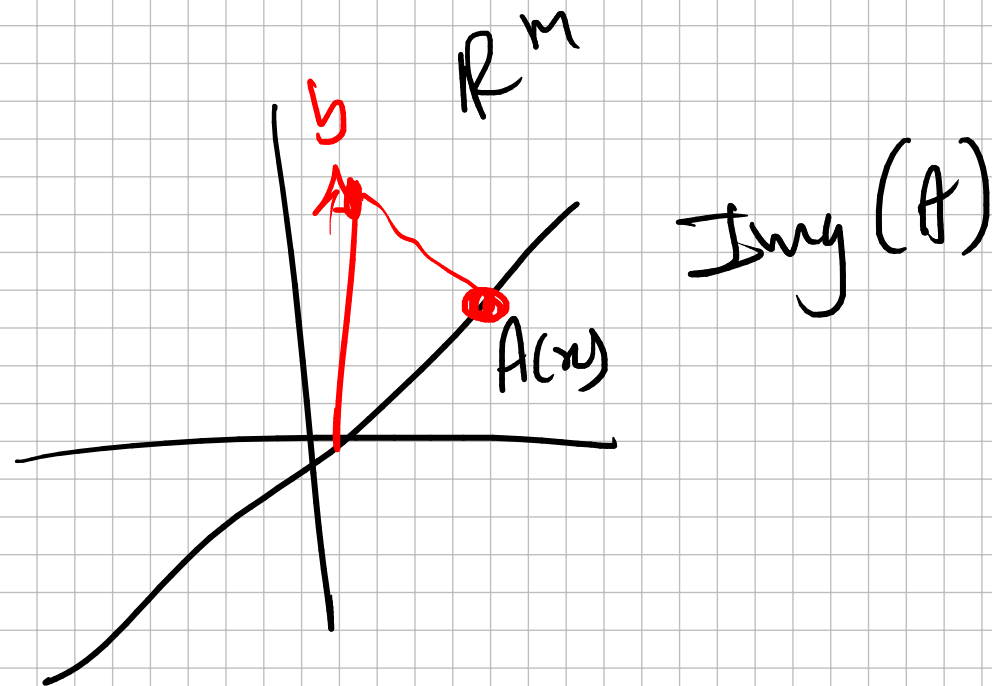
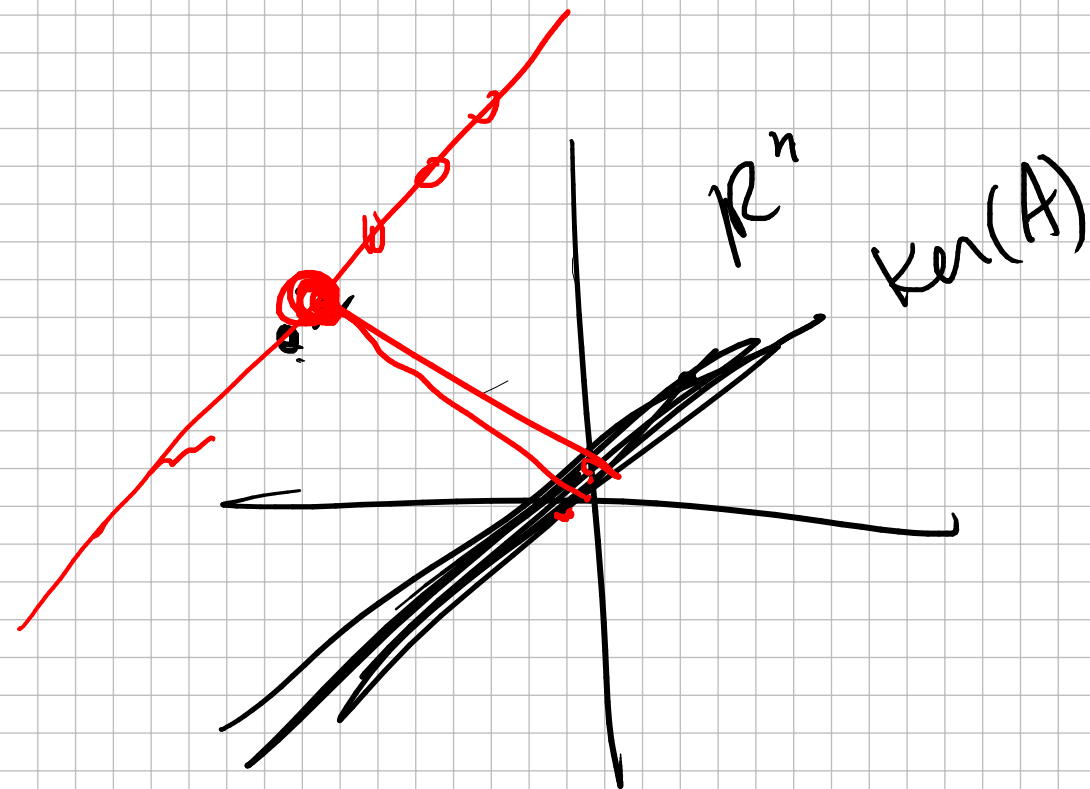
$$x^* = \underbrace{z_1 R_{11}^{-1} Q_1^T b}_{\text{least squares}} + \underbrace{z_2 \alpha_2}_{\text{least squares}}$$

$$\|x^*\|^2 = \|z_1 R_{11}^{-1} Q_1^T b\|^2 + \|z_2 \alpha_2\|^2$$

$\|x^*\|$ minimum

$$x^* = z_1 R_{11}^{-1} Q_1^T b$$

Residual minimum
norm minimum



• Pseudo inversa.

• SVD → Demostración → mínimos usando SVD.

• Factorización de Schur ~~BNS~~.

Simétricas

↓
Diagonalizables

→ Power Iteration

→ Iteración ortogonal
↓